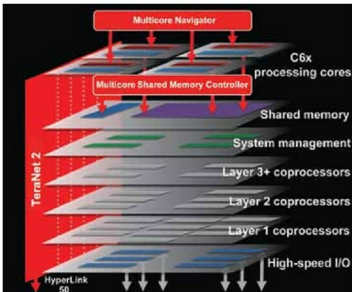


core too, but only at a higher frequency. Rajesh Pathak, advanced technology specialist from Delhi, shares, "Increase in the frequency always comes with a cost of increase in the power consumption. The increased power consumption causes more heat dissipation, resulting in less reliability and more cost. Also, increase in the frequency does not increase the performance by the same scale as parallel instruction execution and pipelining due to slower on/off on the chip's I/Os. Multicore can provide performance enhancement without the need to increase the frequency and also without compromising on power consumption."

Thronsdon believes, "Simply increasing the clock frequency doesn't necessarily translate into improved performance due to bottlenecks in accessing the rest of the system. Also, improving the per-clock-cycle performance beyond a certain point can push single-core design complexity far beyond cost-effective performance scaling." At some point, however, it becomes easier to add a second core to almost double the performance.

Multicore chips have the ability to scale the available processing horsepower more efficiently and dynamically to the current performance requirements in an application. For example, Thronsdon shares, "The peak performance requirement for a Web-connected TV can benefit from running on a multicore platform at top speed. However, for sub-modes of operation, such as use of the TV for simply watching TV in a more traditional manner, one or more cores in the multicore system can be powered off to dramatically decrease power consumption and scale the performance delivered to match the software load."



TI's multicore SoC architecture (Image courtesy: Texas Instruments)

He adds, "Multicore facilitates a 'cache-coherent' SMP platform that provides dynamic load balancing of a variety of system functions that can utilise the available multiple cores in the system-on-a-chip (SoC) more efficiently."

Basically, multicore can be of two kinds. Explains Peter Bushell, director and senior consultant at Software Integrity, "One is the distributed multiprocessing or 'horses for courses,' where a digital signal processor (DSP) works with a general-purpose microcontroller or two non-identical processors are fused together. This is more efficient, cost-effective and reduces power consumption in many cases. The other kind is where several identical processors communicate over some proprietary bus arrangement. This is wonderful for supercomputers and other applications requiring very fast signal processing."

Bushell believes that bringing the second type to the embedded space is simply a push of the technology by chip manufacturers. Indiscriminately shifting the entire field away from simple, cost-effective, low-power solutions towards a complicated multicore hardware (which requires a complicated operating system) may be counterproductive in certain applications.

Other parallel processing devices

The biggest advantage of multicore is that it enables parallel computing. But there are other devices like field-programmable gate arrays (FPGAs), complex programmable logic devices and graphics processing units (GPUs) too that provide parallel processing capability. So can these devices replace multicore in embedded systems?

All the technologies have their own trade-offs, benefits and drawbacks for particular application segments.

Kumaran shares, "GPUs and FPGAs can be used to provide excellent performance-per-watt ratios. However, these devices require specialised programming and specialised toolsets to program. The benefit of homogeneous multicore is a common programming environment, operating systems and hardware support. For certain applications, heterogeneous multicore devices may make better economic sense; e.g., lower cost per unit than homogeneous chipsets."

For graphics processing, "Multicore brings along the simplicity in programmability. When you use a multicore for graphics instead of a GPU, the software development is much easier and flexible," says Subind Kumar, head-Networking and Multimedia Group, Freescale Semiconductor.

Panikkar informs, "FPGAs were not originally designed for floating-point math operations. The software was written for a particular operating system. This purely digital logic hardware approach naturally has limitations and, in fact, the newest generations of FPGAs contain hardened processor cores within them. It ends up being more expensive to do complex math on an FPGA than on a processor and hence multicore still has its place."

GPUs are most suited for data-